Reference Event 1: The 911 Outages After the Derecho of June 2012

Characterization Against the Emergency Services Sector Design Basis

DBA-ESS-RE-1 — Reference Event Instrument, DBA-ESS Series

Lineage: Issued under the Emergency Services Sector Framework v0.7; DBA-MA Parent Framework; conferred on the proposal of Public-Safety Answering and Dispatch (DBA-ESS-AD-FC1)

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Timothy E. Roxey

Eclectic Technologies

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the loss of 911 service across six states after the derecho of 29 June 2012 back through the Emergency Services Sector Design Basis, as the Emergency Services Sector Framework states it and as DBA-ESS-AD-FC1 (*Public-Safety Answering and Dispatch*) applies it to the 911 centers that answer emergency calls. It asks which provisions of the class bore on the event, how each performed, and what the class did and did not correctly draw from it.

The event is the power case of the answering class, and the one from which the class took its finding that a 911 center can be intact, powered and staffed and still receive no calls. Most of what follows confirms the class's reading. The findings that go further concern what the area must know of the networks upstream of its centers, and must have evidence of before the storm: whether the backup power that serves them has been shown to run, whether the paths that carry their calls are as diverse as they are recorded to be, and whether the people who watch those paths can still see them when the power fails.

1.2 Position and Scope

This is a reference event instrument in the sense of the series convention: it reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for the class and framework to disposition. It was issued under the Emergency Services Sector Framework v0.7, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1 at v0.4 on the proposal of *Public-Safety Answering and Dispatch*. Reference events are numbered by conferral, so this instrument carries the first number although it was the tenth of the series to be written.

The scope is set on the author's decisions of 7 October 2026. The event is the loss of 911 service from the storm's passage on the night of 29 June 2012 to the full restoration of service in Northern Virginia on 3 July, anchored on Fairfax County and its neighbors, where the centers were intact and the carrier's central offices failed, with West Virginia as the second case, where several centers failed at the center itself. The federal rule that followed, and its revision in June 2026, are recorded as the regulatory sequel. The carriers are named as the regulators' findings name them, each finding is attributed to the body that made it, the carriers' responses are stated beside the findings, and this instrument makes no finding of fault of its own. Individuals are named by role.

The primary sector is Emergency Services, since the consequence at issue is the loss of the public's ability to reach 911. The failures that caused most of it lay on the communications side of the boundary the class draws (§8.3), in carrier central offices, 911 circuits and location links; Communications is the impacted sector, and Energy the sector whose failure began the event. The constructs applied are defined in the Foundational Definitions and cited, not restated: FD-BL

(*The Bright Line*), FD-BR (*The Broker*), FD-LD (*Layer Decomposition and Determinism*), FD-LS (*Local Sovereignty*), FD-BL-D1 (*Mode of Discharge*) and FD-BL-D2 (*Indication Integrity*). FD-REG holds their versions.

2. Sources and Attribution

2.1 Sources and Their Limit

The principal source is the report of the Federal Communications Commission's Public Safety and Homeland Security Bureau, *Impact of the June 2012 Derecho on Communications Networks and Services: Report and Recommendations*, of January 2013, which gives the outage by state and by center and the failures in the carriers' central offices. Beside it stand the final report of the staff of the Virginia State Corporation Commission of 17 January 2013; the Metropolitan Washington Council of Governments' *Final Report of 9-1-1 Service Gaps During and Following the Derecho Storm*, released 13 March 2013; the Commission's order *Improving 911 Reliability* of 12 December 2013 (FCC 13-158), its further notice of March 2025 (FCC 25-21) and its order of June 2026 (FCC 26-39); and Verizon's public statement of January 2013. Contemporaneous reporting by Connection Newspapers, Nextgov and Potomac Local carries the account of the Northern Virginia centers. No non-public material is used.

The structural facts, the timeline in Northern Virginia and the scale are established by the Bureau's report and are not contested; the cause of the central-office failures and whether any center's own failure was avoidable are attributed findings, stated in Section 2.2. One limit is stated plainly. The Bureau's report and the Council of Governments' report were read through releases and summaries of their text rather than page by page, and the outage durations for the twenty Virginia centers that lost service in part are not given in the material read. OI-2 registers the full reading as owed, and no finding rests on a figure that reading could move.

2.2 Attribution

The Commission's Bureau found that in most cases the 911 problems could and would have been avoided, and that the carriers' implementation of established best practices was neither as diligent nor as consistent as needed. The Virginia Commission's staff found the outage avoidable, caused by Verizon's failure to maintain its central offices and made worse by its inability or failure to monitor and respond. The Council of Governments named the power loss, the failure of one of the two backup generators at Verizon's central offices, and improper maintenance and incident response. Verizon said it had resolved the causes of the generator failures, undertook voluntary audits of backup power in Virginia, Maryland and the District of Columbia, and adopted recommendations of the region's 911 directors. This instrument states each finding as the body's that made it and the response as Verizon's, and makes no finding of fault. Its findings do not depend on one: on every account the centers in Northern Virginia were powered and staffed, and the calls stopped upstream of them.

The Commission's order of December 2013 says the record does not reflect avoidable failures on the part of 911 centers. Its Bureau's report of the same year records that eight of the twenty-seven affected centers in West Virginia lost service because of issues at the center itself, and that backup generators at several centers failed to start. This instrument states both and treats the second as the more specific account of what happened in West Virginia.

The record of the event is closed; the rule it produced was revised in June 2026, which F-5 takes up.

3. The Event

On the night of 29 June 2012 a derecho, a long-lived windstorm, cut commercial power across the Mid-Atlantic, in some places for nearly two weeks. In Northern Virginia two of Verizon's central offices failed. At Arlington commercial power was lost at 10:55 p.m. One of the two backup generators would not start, because air had entered its fuel system, and the other shut down under the load; the batteries carried the office until about 5:00 a.m. on 30 June. The uninterruptible power supply feeding the company's monitoring of its network lasted thirty minutes, and the Bureau found that Verizon lost the ability to monitor its network in Northern Virginia. Load testing of the office's generators had been suspended since at least 2011. At Fairfax commercial power failed at 10:35 p.m. A generator did not start because of a malfunction in its automatic start; a technician who arrived at about 7:30 a.m. did not check it, and by about 12:15 p.m. had noticed the problem and restarted it by hand.

Fairfax County's 911 center, serving more than 1.1 million people, was operating on its own generators throughout. At 7:36 a.m. on 30 June 911 calls stopped arriving, and for the next seven hours no 911 call was completed to the center. Service was fully restored at 11:30 a.m. on 3 July. Nearly 1,900 calls to 911 entered Verizon's network in the first twenty-nine hours and were not routed to the county. The Bureau found that the center's redundancy was compromised by transport failures in the Fairfax office and other single points of failure in Verizon's network, which a diversity audit might have identified. The center learned of the trouble when its phones stopped ringing; the carrier had sent designated county staff an e-mail about the Arlington office, and did not telephone. Verizon did not know that Fairfax's 911 service was down until the county told it.

The neighbors fared alike. Prince William County found by 8:07 a.m. that it was receiving no 911 calls; most calls were delivered by the morning of 1 July without the caller's number or location, which returned early on 2 July, and the county told the public through the media and its alerting systems to use its non-emergency numbers. Manassas Park took six hours to reach Verizon, which did not know of the outage. In Manassas the overflow and administrative lines were down with the 911 lines. Arlington County never lost service entirely: at about 5:00 a.m. three of its four location links failed, all three running through one central office, and at about 9:40 a.m. the failure of the Fairfax office took down half of its 911 trunks. Twenty further Virginia centers lost service in part.

In West Virginia, served largely by Frontier, eleven centers lost service for periods from less than an hour to more than twelve hours. A generator in one carrier office failed after about twenty-two hours, and the company lost the monitoring of its network after a fiber cut, having disconnected the backup circuit a week before. Most of the affected centers made the first call to the carrier to report their outages. Eight of the state's twenty-seven affected centers lost service because of issues at the center itself, and backup generators at several centers failed to start. In Ohio at least two carrier backup generators failed to start.

Across six states seventy-seven 911 centers serving more than 3.6 million people lost some connectivity, and seventeen centers in three states, serving more than two million, lost service completely. In December 2013 the Commission required covered 911 service providers to take reasonable measures for the diversity of 911 circuits, backup power at the central offices that directly serve 911 centers, and diverse network monitoring, and to certify each year that they had done so; 911 centers were left outside the rule.

4. Technical Decomposition

The event decomposes into initiating elements (I-n), which made the loss of 911 likely before the storm arrived, and effect elements (E-n), which carried it to the people who could not reach help.

The initiating elements are three: backup power that had not been tested (I-1), diversity recorded and not present (I-2), and monitoring that shared the power of what it watched (I-3). The effect elements are five: an intact center with no calls (E-1), recognition from the center's side (E-2), calls without location (E-3), the public told to call another number (E-4), and centers that failed themselves (E-5). Table 4-1 maps each to the provisions it engaged and the finding it supports.

Composed, the elements describe a forecastable kind of event, a regional loss of power, meeting backup power that had not been tested, diversity that was recorded and not present, and monitoring that went dark with the network. No element was exotic. Their composition is what *Public-Safety Answering and Dispatch* calls chain A, the loss of answering, from a cause upstream of the center, at the reach of six states.

Table 4-1. Event elements and their mapping. Section references are to Public-Safety Answering and Dispatch v0.2.

Ref	Description	Provisions engaged	Finding
I-1	Arlington generators not load-tested since 2011; failed to start or carry load; Fairfax auto-start failed; center generators failed in West Virginia	§4.1, §5.6; Access and Edge	F-2
I-2	Fairfax circuits with single points of failure; Arlington's location links through one office	§3.4, §4.2; CDB-4	F-3
I-3	Carrier monitoring on a thirty-minute supply; backup monitoring circuit disconnected	§4.2, §7.2; CDB-4	F-1
E-1	Fairfax intact, no completed 911 call for seven hours; 1,900 calls not routed in 29 hours	§3.2, §4.2	F-1

E-2	Centers recognized the failure by silence and told the carrier; e-mail rather than call	§3.2, §4.2, §7.3 (CDB-4, CDB-5)	F-1
E-3	Calls delivered without number or location, in Prince William until early 2 July	§3.2, §4.2	F-4
E-4	Public told to use non-emergency numbers	§4.2, §5.7	F-4
E-5	Eight West Virginia centers failed at the center; generators did not start	§4.1, §5.6, §5.8	F-2

5. Tier Determination

Against the four characteristics of military action, as *Public-Safety Answering and Dispatch* §5.2 states them for this class: intentionality is absent, the storm being weather and no account holding that anyone intended the loss of power or of 911; persistence is absent, the conditions that made the loss likely having persisted for months or years by neglect, not design; precision is absent, the failures concentrating on two central offices because those offices served the 911 centers and their generators failed, not because anything selected them; and denial is absent as a characteristic and present in part as a consequence, the loss of monitoring denying the carrier the sight of its own network.

The event does not meet the threshold for the military-action extension. All four characteristics are absent. It is a non-adversarial event of the class's first and second design-basis categories, the loss of commercial power outlasting backup and the failure of call delivery upstream of the center (*Public-Safety Answering and Dispatch* §4.1, §4.2), in chain A, at the reach of six states. The provisional placement the author set is confirmed. That is the tier determination returned for the DBA Reference Event Register.

What a directed actor could do through the same path (strike the grid where the central offices that carry an area's 911 circuits depend on untested backup) runs forward from the event rather than back through it. It is offered to Project 9 as material for the class's compound military-action scenario (§5.3) and is not developed here. Its bearing on this reading is recorded in Findings F-2 and F-3.

6. Characterization Against the Design Basis

Section references are to *Public-Safety Answering and Dispatch* v0.2.

6.1 §3.2 The Two Governing Measures

Held. The class counts alternate-answering transfer time from the caller's side and bounds it by recognition time. Fairfax's center was intact throughout; the transfer time was set by the time to recognize that calls had stopped, to reach the carrier and to have calls rerouted, and for Manassas Park the time to reach the carrier alone was six hours. Operate-through duration counts only

power and staff on hand unless their supply is assured, and the Fairfax center, on its own generators, held throughout.

6.2 §3.4 The Alternate-Routing Map

Held, with a gap. The map records where each area's calls go when its center cannot take them, what invokes each transfer, what each alternate shares with its primary, and the population whose calls pass through each upstream element. The derecho shows that the record of what is shared must be checked against the physical routes, since the diversity the area believed in was not present (F-3), and that the map does not record the backup power of the upstream elements it depends on, though that is what failed (F-2).

6.3 §4.1 Loss of Commercial Power Outlasting Backup

Held, with a gap. The class requires each center, backup center and radio site to carry backup power at least equal to the area's design duration, protected against the event that takes the grid. Fairfax's center met it; several West Virginia centers did not, their generators failing to start; and the carrier offices that failed in Virginia, outside the class, failed the same way. The class requires backup power to exist and to be protected. It does not say that backup power is credited only when it has been shown to start and carry its load, at an interval and by a test that would have revealed the failures of the derecho (F-2).

6.4 §4.2 Failure of Call Delivery Upstream of the Center

Held, with a gap. The class requires the center to recognize from its own records that calls have stopped, requires whoever operates an element in the call path to detect the failure of call completion at a severity set by the consequence, requires an alternate path that does not share the failed element, and requires the public to be told how to reach help. Each answers an element of the event. What the class does not say is that the operator's detection must keep working when the element it watches loses power. Verizon's monitoring lasted thirty minutes and Frontier's backup monitoring had been disconnected (F-1).

6.5 §3.2 and §4.2 Calls Without Location

Held, with a gap. The class counts transfer time until calls ring at a staffed position with the caller's location delivered, so Prince William, Manassas Park and Arlington were still in transfer while calls came back without location. That is right. What the class does not say is how a center works in that state, answering and finding the caller without the system's help (F-4).

6.6 §5.6 and §5.7 The Surviving Answering Condition and Its Residuals

Held. The class's first declared residual is a failure of call delivery upstream of every center, for which the response is public instruction by channels that do not depend on the telephone network. The derecho realized that residual for seven hours in Fairfax and for more than a day in

Prince William and Manassas, and the centers' instruction to use non-emergency numbers was the response the class names. Non-emergency lines that run through the same failed office are no alternative, which F-4 takes up.

6.7 §7 The Communications Design Basis

Held. The class's finding that a silent center looks like a quiet night (§7.2) is this event as much as the 2014 outage. CDB-4 requires each center to learn by a means independent of the delivery network that calls have stopped and to reach its providers over a channel that shares nothing with its primary links; CDB-5 requires the centers and the carriers to coordinate rerouting when their usual channels are down. The e-mail, the six hours to reach the carrier and the centers that told the carrier of its own failure are each a requirement unmet. The class's reading needs no change; F-1 asks that the operator's notification be made independent of the operator's own power.

6.8 §8.4 The Regulatory Context

Held, with a gap. The failures lay on the communications side of the boundary the class draws (§8.3), and in 2012 the obligations the derecho shows were missing were not the 911 authorities' to enforce. The Commission's rule of December 2013 placed them on the providers by annual certification, and the class's §8.4 describes that rule. The class does not rest on the rule, but an area that relied on the annual certification as its evidence of what lies upstream has lost a yearly source of it now that the certification is to be one-time (F-5).

6.9 The AI-Governance Design Basis

Not exercised. No machine-learning or generative system bore on the loss of service. The class's §6.4 places automatic failover of call routing as an action on its consequence, authorized in advance under FD-BL-D1 (*Mode of Discharge*) on a measured trigger such as the completion rate. The derecho is a case for that trigger: Fairfax's completion rate fell to nothing at 7:36 a.m. and no one who could reroute the calls saw it. A failover rule keyed to the measure would have acted on it, provided the measure itself survived the loss of power, which F-1 requires.

6.10 Summary of Design Basis Performance

Of the nine readings, §3.2, §5.6 with §5.7, and §7 Held; §3.4, §4.1, §4.2, the calls-without-location reading of §3.2 and §4.2, and §8.4 Held, with a gap; no provision Missed; and the AI-governance design basis was Not exercised. Where the class falls short, it does so on what the area must have evidence of upstream: that the operator's detection survives the loss of power, that backup power is credited only when shown to run, that diversity is verified against the physical route, how a center works when calls arrive without location, and what evidence of the upstream elements the area holds now that the federal certification is a one-time act.

7. Findings

Section references are to *Public-Safety Answering and Dispatch* v0.2 unless another instrument is named. Two findings share ground with DBA-ESS-RE-2 (*The Multistate 911 Outage of April 2014*), which is the home of the shared findings: F-1 here adds to its F-3, and F-3 here adds to its F-5.

7.1 F-1 — Detection of a Silent Failure Must Survive the Loss of the Power It Watches

The class requires the operator of each element in the call path to detect the failure of call completion through it, at a severity set by the consequence, and to notify the centers on that detection (§4.2, CDB-4); where that detection is placed and how it is rated is the finding of *The Multistate 911 Outage of April 2014* F-3, and is not repeated here. The derecho adds what that finding does not: the detection must survive the loss of the element's power. The carrier's Northern Virginia monitoring ran on a supply that lasted thirty minutes, and in West Virginia the backup monitoring circuit had been disconnected. The carrier did not know Fairfax had lost 911 until the county told it; the county learned from an e-mail and the silence; and Manassas Park took six hours to reach the carrier. The class should state that the detection and notification it places on the operator of an element are powered and connected independently of that element, for at least the area's design duration; that notification of a center is made by a means the center will receive at once, not by an e-mail to designated staff; and that the center's own recognition of silence, which worked, remains required because the operator's may fail. Affected: §4.2, §7.3 (CDB-4, CDB-5); *Access and Edge*; *Core Switching, Routing and Signaling*. Disposition: R-1, R-7.

7.2 F-2 — Backup Power Is Credited Only When It Has Been Shown to Run

The class requires backup power at least equal to the design duration and protected against the event that takes the grid (§4.1), and counts operate-through duration on what is on hand (§3.2). The generators that failed in Verizon's Arlington office had not been load-tested since at least 2011; one in its Fairfax office failed in its automatic start; carrier generators in Ohio did not start; and generators at several West Virginia centers did not start. The class should state that backup power is credited toward operate-through duration only when it has been shown to start automatically and to carry its full load for a stated period, at a set interval and by a test that would reveal the failures of the derecho; that a generator not so shown counts as no generator; and that the alternate-routing map records, for each upstream element whose loss would stop the area's calls, the evidence of its backup power on the same terms. The carriers' central offices are *Access and Edge*'s, and the test there is that class's to state. Affected: §3.2, §3.4, §4.1, §5.8; *Access and Edge*. Disposition: R-2, R-7.

7.3 F-3 — Diversity Is Verified Against the Physical Route

The class requires an alternate routing path that does not share the failed element and a map that records what each alternate shares with its primary (§3.4, §4.2); that a shared design counts as a shared element is the finding of *The Multistate 911 Outage of April 2014* F-5. The derecho adds that what is recorded as not shared must be checked against the ground. The circuits that served Fairfax had single points of failure that the Bureau found a diversity audit might have revealed, and Arlington's three location links ran through one office. Diversity recorded and not present is no diversity. The class should state that the shared elements recorded in the map are verified against the physical route of each circuit, including the links that deliver the caller's location, at a set interval and on any change to the network; and that location links are mapped with the same care as the voice path, because an area whose calls arrive without location has not completed its transfer. Affected: §3.4, §4.2; CDB-4. Disposition: R-3.

7.4 F-4 — The Degraded States Between Silence and Service Are Planned

In Prince William, Manassas Park and Arlington calls returned before their location data, and in Prince William arrived without the caller's number or location until early on 2 July. Meanwhile centers told the public to call non-emergency numbers, which in Manassas were down with the 911 lines. The class counts these areas as still in transfer, which is right, and requires the public to be told how to reach help (§3.2, §4.2). The class should state that each center plans for the states between silence and full service (calls without location, calls on administrative lines, calls rerouted to a neighbor without the area's unit picture) with procedures for each, among them call-back and verbal location; and that any number the public is told to call instead of 911 is verified, before it is announced, not to run through the element that has failed. Affected: §3.2, §4.2, §5.7. Disposition: R-4.

7.5 F-5 — The Area Holds Its Own Evidence of What Lies Upstream

The Commission's rule of December 2013, which the class's §8.4 describes, required providers to certify each year that they had taken reasonable measures for circuit diversity, central-office backup power and diverse monitoring. In June 2026 the Commission decided to replace the annual certification with a one-time certification updated on material change, to be filed after an eighteen-month transition. An area that took the certification as its evidence has lost a yearly source of it. The class should state that the 911 authority holds, in its alternate-routing map, its own evidence of the diversity, backup power and monitoring of each upstream element whose loss would stop its calls, on a cadence it sets; that where the contract test of §8.3 places a function on the authority's side, the contract binds the provider to supply that evidence; and that §8.4 records the rule as it now stands. Affected: §3.4, §8.3, §8.4. Disposition: R-5.

7.6 F-6 — Withdrawn at v0.2

Its substance, correcting the class's own account of the event at *Public-Safety Answering and Dispatch* §2.3, is a correction to another instrument rather than a test of the design basis, and is withdrawn under the Reference Event Compression Standard. The class has already taken the correction in, at its v0.3, where §2.3 cites this finding, so no erratum is owed. Its return of the upstream findings to the communications classes is carried by R-7, and its naming of the impacted sectors by R-8. The correct populations, durations and West Virginia account are held in Section 3.

8. Recommendation Register

Priority is stated on the series' one-to-three scale, where 1 is the most operationally consequential. R-1 to R-5 dispose to the next revision of *Public-Safety Answering and Dispatch*; R-7 disposes to *Access and Edge* and *Core Switching, Routing and Signaling*; R-8 disposes to the register.

ID	Recommendation	Affected	Direction	Pri
R-1	Make detection survive the loss of power	§4.2, §7.3	State that the operator's detection and notification are powered and connected independently of the element watched, for the design duration; that notification reaches a center at once, not by e-mail; and that the center's own recognition of silence remains required.	1
R-2	Credit backup power only when shown to run	§3.2, §3.4, §4.1, §5.8	Credit backup power toward operate-through duration only when shown to start automatically and carry full load for a stated period, at a set interval; count an untested generator as none; record the same evidence for each upstream element in the map.	1
R-3	Verify diversity against the physical route	§3.4, §4.2	Verify the shared elements recorded in the map against the physical route of each circuit, including location links, at a set interval and on change; map location links as carefully as voice.	1
R-4	Plan the degraded states	§3.2, §4.2, §5.7	Plan for calls without location, on administrative lines and rerouted without the unit picture, with call-back and verbal location; verify any substitute number before announcing it.	2
R-5	Hold the area's own evidence of the upstream elements	§3.4, §8.3, §8.4	Hold in the map the area's own evidence of diversity, backup power and monitoring of each upstream element on a set cadence; bind the provider to supply it where the contract test places the function on the authority's side; update §8.4 to the rule of June 2026.	2
R-6	Withdrawn at v0.2	§2.3	Its substance, correcting the class's account of the event, was taken in by Public-Safety Answering	—

			and Dispatch at its v0.3; withdrawn with F-6.	
R-7	Return the upstream findings	Access and Edge; Core Switching, Routing and Signaling	Record for each class's next revision Findings F-1 to F-3: monitoring independent of the element's power, central-office backup power credited only when tested, and location-link diversity verified against the route.	2
R-8	Name the sectors	DBA Reference Event Register	Ask that the register name Communications as the impacted sector and Energy as the sector whose failure began the event. Entered by Project 0 on 7 October 2026.	3

9. Open Items

OI-1, the register row, is closed: Project 0 entered the tier determination of Section 5, Communications as the impacted sector and Energy as the sector whose failure began the event, the instrument version and the counts on 7 October 2026. OI-2 stands: the Bureau's report and the Council of Governments' report are owed a full reading, which should settle the outage durations of the twenty Virginia centers that lost service in part, the Council of Governments' own account of call volumes and the centers' workarounds, and the exact wording of the Commission's rule as revised in June 2026; no finding rests on a figure that reading could move. OI-3 is closed: *Public-Safety Answering and Dispatch* took in R-1 to R-6 at its v0.3, including the correction of §2.3 and the conformance of §10.5, and records R-7 as returned to the communications classes. Section 6 remains a reading of the class at v0.2, as issued.

10. Standing of This Document

This instrument establishes where the 911 outages after the derecho sit within the Emergency Services analytical structure, which provisions bore on them, and what the class did and did not correctly draw from its own power case. It does not modify the class or the framework, and makes no finding of fault; the findings it cites are the regulators', stated as theirs.

The derecho and the multistate outage of April 2014 (DBA-ESS-RE-2) are the class's two cases of an intact center that received no calls, and should be read together. In 2014 the cause was a counter in a routing center and the alternate was ready and unused for six hours; in 2012 the cause was power in two offices and the alternates shared them. In both the centers recognized the silence before the operator of the failed element did. DBA-ESS-RE-7 (*The Warning Failure in the Lahaina Wildfire*) found that the alerting authority could not see that its messages were not arriving; the derecho is the same blindness at the 911 center, and F-1 here and its F-1 are one obligation applied to two directions of travel. Under the triage rulings of 9 October 2026 the Lahaina finding moves into DBA-ESS-RE-9 (*Ukraine Air-Raid Alerting*) as a preface, and this cross-reference follows it there when that instrument migrates. The instrument should be revised

on any revision of *Public-Safety Answering and Dispatch* that dispositions these findings, on the full reading of OI-2, or on any further change to the Commission's rule on 911 reliability.

References

Federal Communications Commission, Public Safety and Homeland Security Bureau, *Impact of the June 2012 Derecho on Communications Networks and Services: Report and Recommendations*, January 2013, PS Docket 11-60, the source of the outage by state and center and of the central-office failures. Federal Communications Commission, *Improving 911 Reliability*, Report and Order, FCC 13-158, 12 December 2013; Further Notice of Proposed Rulemaking, FCC 25-21, March 2025; Second Report and Order and Second Further Notice of Proposed Rulemaking, FCC 26-39, adopted 25 June and released 26 June 2026. Virginia State Corporation Commission, staff final report on 911 service outages after the 29 June 2012 derecho, Case PUC-2012-00042, as released 17 January 2013. Metropolitan Washington Council of Governments, *Final Report of 9-1-1 Service Gaps During and Following the Derecho Storm*, as released 13 March 2013.

Verizon, statement on strengthening 911 service, January 2013. Nextgov, "FCC slams Verizon for 9-1-1 failures," 10 January 2013. Connection Newspapers, "Complete Failure," 30 August 2012, the source of the center director's account. Potomac Local, report on the Virginia Commission staff's findings, 18 January 2013.

Revision History

Version 0.1, October 2026: issued in Project 9 under the Emergency Services Sector Framework v0.7, for the event conferred as DBA-ESS-RE-1 on the proposal of *Public-Safety Answering and Dispatch*. On the author's decisions of 7 October 2026: Northern Virginia as anchor and West Virginia as second case, from 29 June to 3 July 2012, with the federal rule and its revision of June 2026 as sequel; carriers named as the regulators' findings name them, each finding attributed and Verizon's response stated beside them, with no finding of fault; tier set provisionally and confirmed. An independent check corrected the record before issue. Findings F-1 to F-6; recommendations R-1 to R-8.

Version 0.2, 9 October 2026: migrated to Project 11 under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), compressed under the Reference Event Compression Standard v0.3 and brought to house style. Cut: the general description of the reference event instrument; the restatement of the governing instruments (now citations); the separate evidentiary-class and currency sections (merged into Section 2); the event narrative and the decomposition, tightened; the four characteristics, merged into one paragraph; the §8.3 boundary narrative, with §8.4 kept; the separate relationship and maintenance sections, merged into Section 10; the list of DBA instruments in the references, now cited in the text; and the constructed analog of Section 5.6, which runs forward and is offered to Project 9 as use-case material. Section 6 now uses the standard's verdicts, and the separate account-of-the-event

subsection is removed with F-6. Overlaps with DBA-ESS-RE-2 are resolved by cross-reference under the triage of 9 October 2026: RE-2 is the home of the detection-placement finding (its F-3) and the shared-design finding (its F-5), and F-1 and F-3 here now cite them and keep only what this event adds, detection that survives the loss of power and diversity verified against the physical route; no finding moved and no number changed. Stubs created: F-6 and R-6, whose correction of *Public-Safety Answering and Dispatch* §2.3 the class had already taken in at its v0.3, so that no erratum is owed, with F-6's other dispositions carried by R-7 and R-8. Unchanged: the designator, title, tier determination, attribution statements, sources, and the substance and numbering of F-1 to F-5, R-1 to R-5, R-7 and R-8. OI-1 closed on the register entry of 7 October 2026, and OI-3 on the class's v0.3. Verified before migration: every finding and recommendation number present at v0.1 is present here as an entry or a stub.